



CS-302 Artificial Intelligence

WEEK 02

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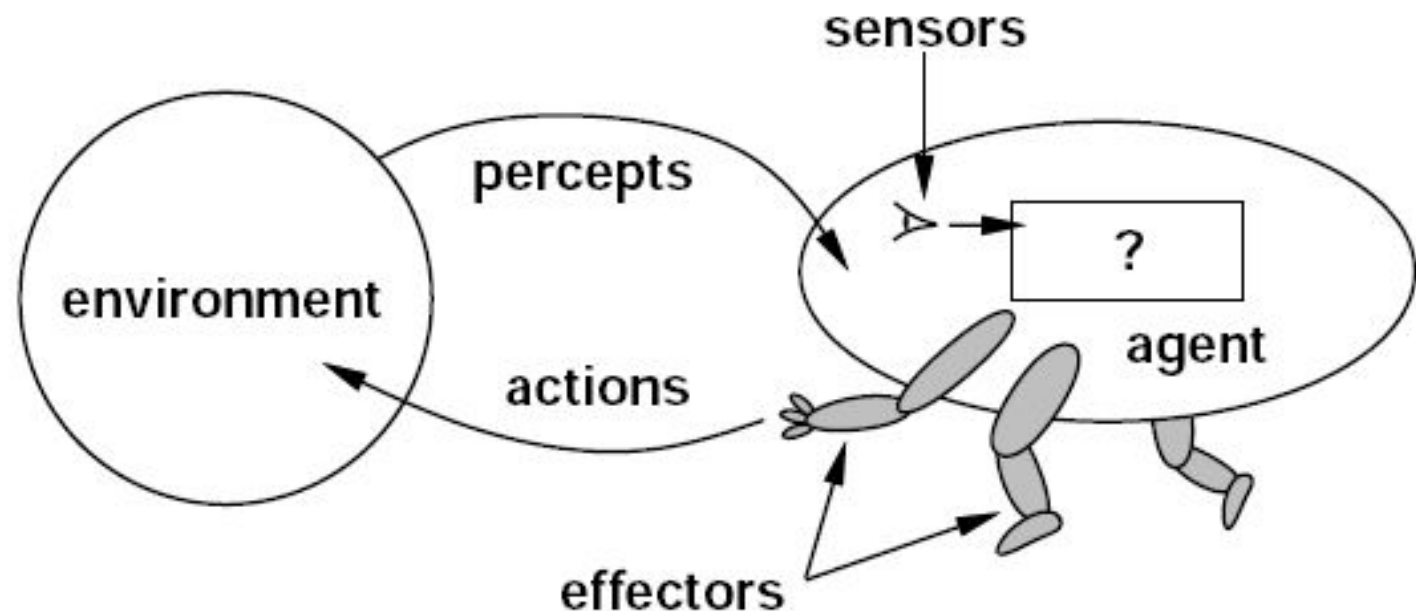
Intelligent Agents

Agents and Environments

An **agent** is anything (an entity) that can be viewed as perceiving its **environment** through sensors and acting upon that environment through **actuators**.

This definition includes:

- Robots, humans, programs



Examples of Agents



Humans

senses

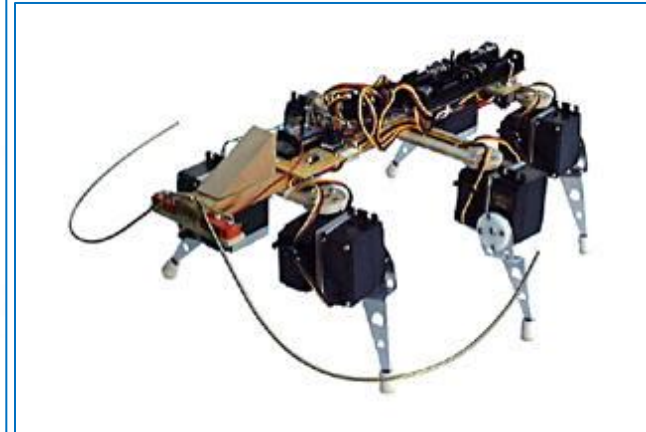
body parts

```
Else
  If (Len(Trim$(str1
    blnFileNext =
  Else
    ' Process data
    intColon = InStr
    If (intColon
      strName =
      strValue =
      strName = S
      Call xnNode
    End If
  End If
End If
```

Programs

keyboard, mouse, dataset

screens, speakers, files



Robots

sensors, cameras,

motors, limbs

Rational / intelligent agents

(review:

lec1)

- **An intelligent agent** is an agent that acts intelligently.
 - Its actions are appropriate for its goals and circumstances
 - It is flexible to changing environments and goals.
 - It learns from experience.
 - It makes appropriate choices given perceptual limitations and finite computation.
- **Important Consideration:** Computational limitations make perfect rationality unachievable.

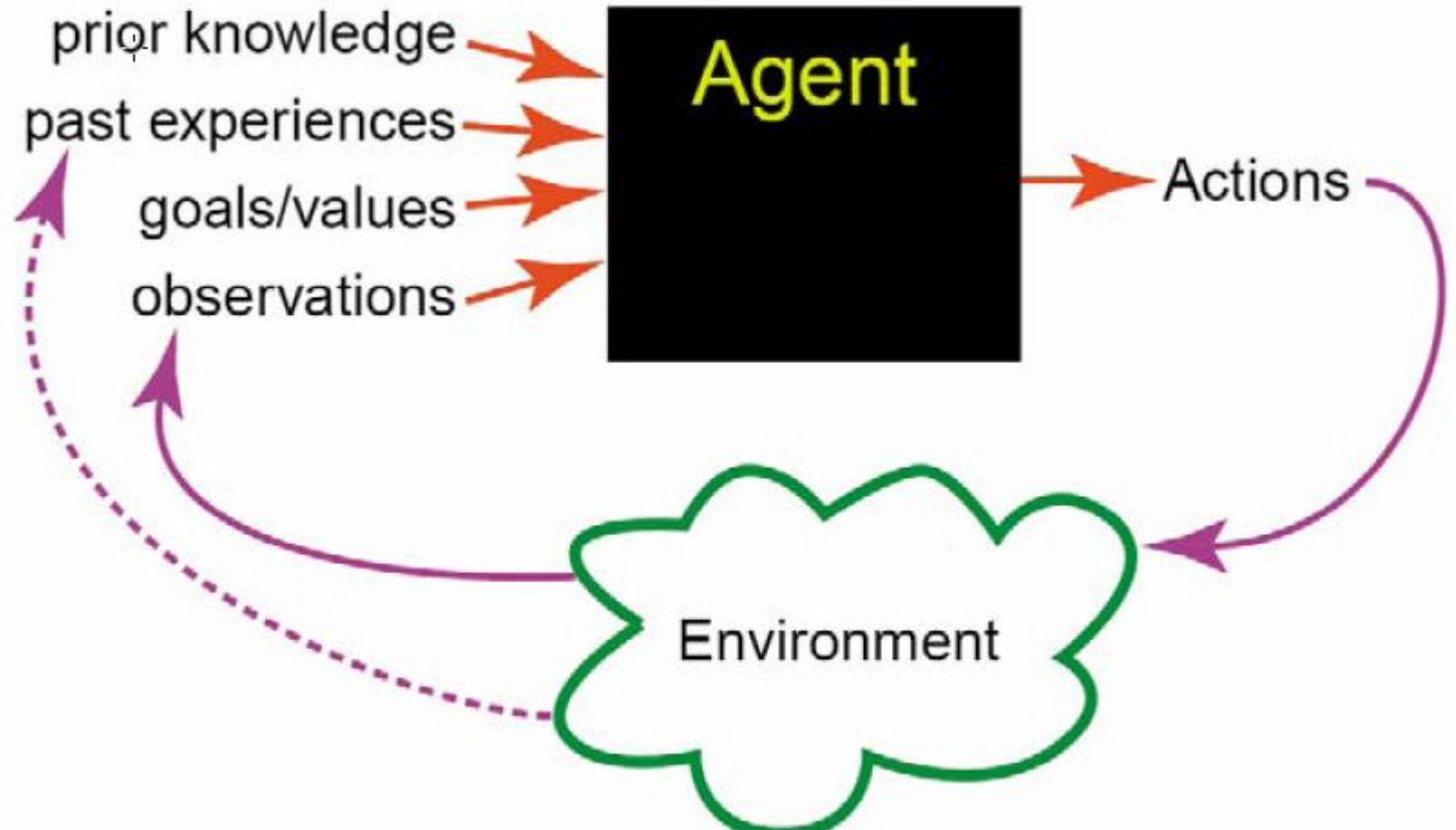
Rational Agents

A **rational agent** is one that does the right thing

- Need to be able to assess agent's performance
 - Should be independent of internal measures
 - A chess AI **may use deep algorithms**, but its rationality is judged based on **how well it wins games**, not how complex its decision-making process is.
- Ask yourself: has the agent acted rationally?
- A rational AI is one that not only has intelligence but applies it efficiently in real-world situations.

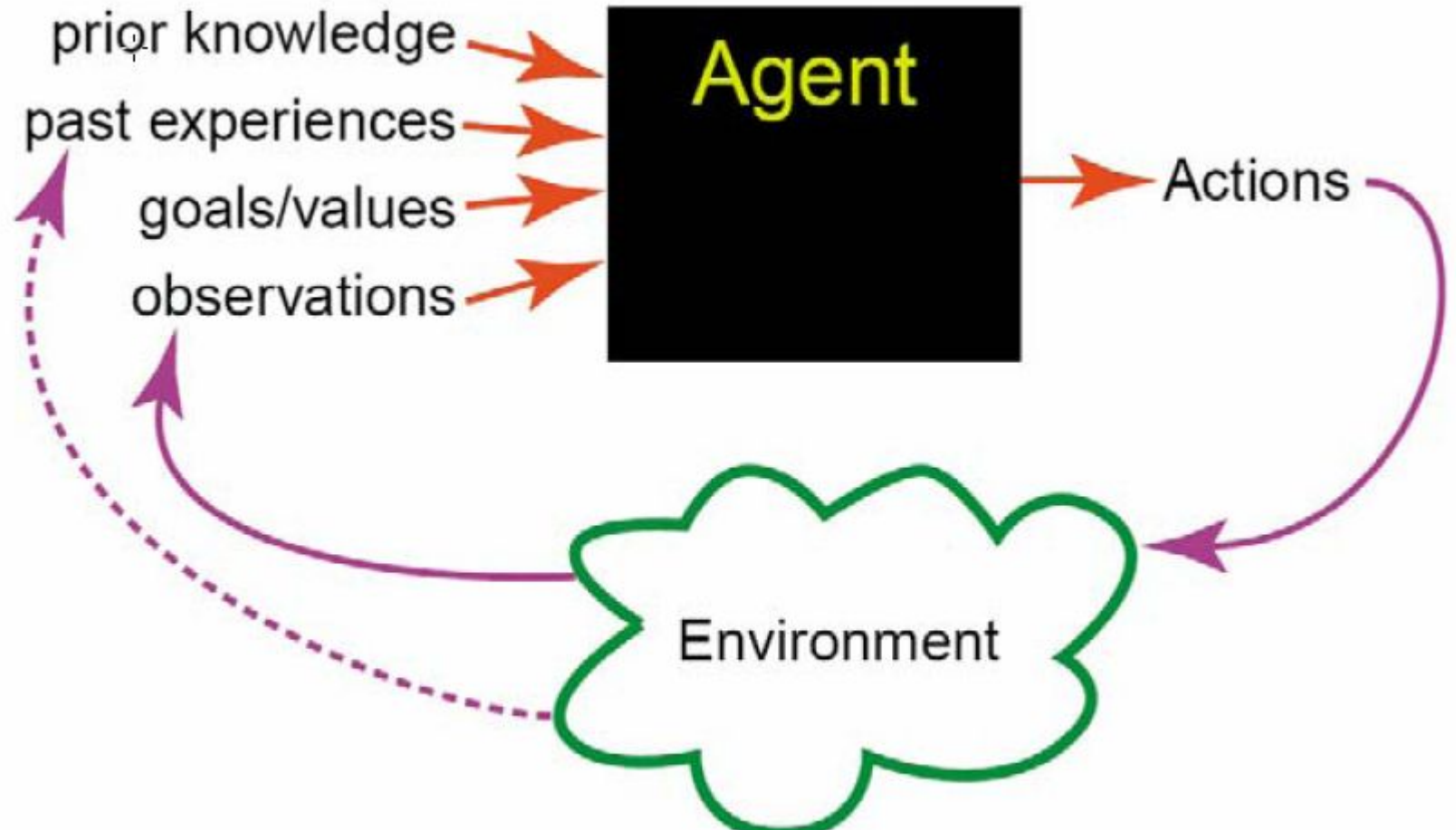
Agents Acting in an Environment

$$f : \mathcal{P}^* \rightarrow \mathcal{A}$$



Agents Acting in an Environment

$$f : \mathcal{P}^* \rightarrow \mathcal{A}$$



Autonomy in Agents

The **autonomy** of an agent is the extent to which its behaviour is determined by its own experience

- Extremes
- No autonomy – The agent completely ignores environmental data and acts only on predefined rules.
 - Example: A traditional calculator that only performs calculations based on predefined logic. It does not learn or adapt.
- Complete autonomy – The agent has no internal knowledge and learns everything from scratch without preprogrammed rules.
 - Example: A newborn baby who has no knowledge of the world and must learn everything by experience.
- Ideal – design agents to have some autonomy
 - Possibly good to become more autonomous in time

Autonomy in Agents

- **Low Autonomy:** A **fixed-rule chatbot** that only gives predefined responses.
- **Medium Autonomy:** A **self-driving car** that learns from new traffic patterns but still follows programmed rules.
- **High Autonomy:** **AI in robotics** that can adapt its behavior completely based on real-world experiences.

Types of Agent Programs

- Simple Reflex Agents
- Model-Based Reflex Agents
- Goal-Based Reflex Agents
- Utility-Based Agents
- Learning Agents

Types of Agent Programs

Simple Reflex Agent

- The simplest kind of agent
- These agents *select actions on the basis of the current percept, ignoring the rest of the percept history.*
- *No memory*
- Implemented through *condition-action rules.*
 - **If** dirty **then** suck.

Types of Agent Programs

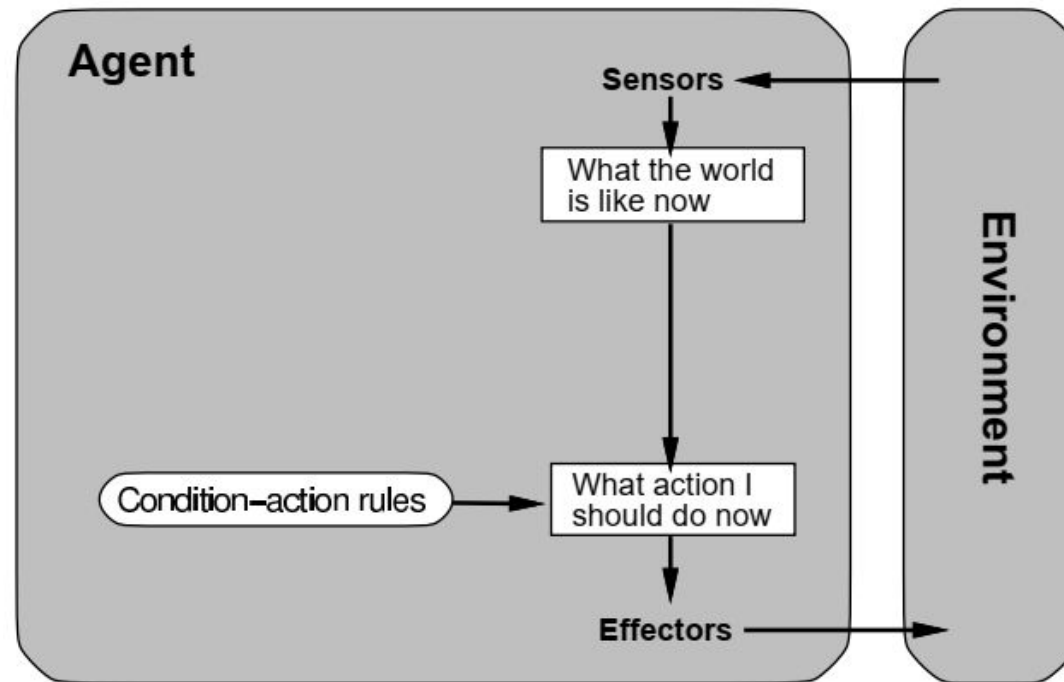
Simple Reflex Agent

- *Characteristics*
 - Very limited knowledge or intelligence
 - Only works if the environment is fully observable.
 - In partially observable environments, they can get stuck in infinite loops because they do not recall previous conditions.
 - Example: A vacuum cleaner that turns in the same direction repeatedly without realizing it's stuck in a corner.

Types of Agent Programs

Simple Reflex Agent

Simplex Reflex Agent: Schema



Types of Agent Programs

Model-Based Reflex Agents

- Unlike Simple Reflex Agents, these agents do not rely only on current percepts.
- Instead, they keep track of unobservable aspects of the environment.
- It builds a model of the world based on its previous experiences.
- This allows the agent to predict or estimate missing details of the environment.

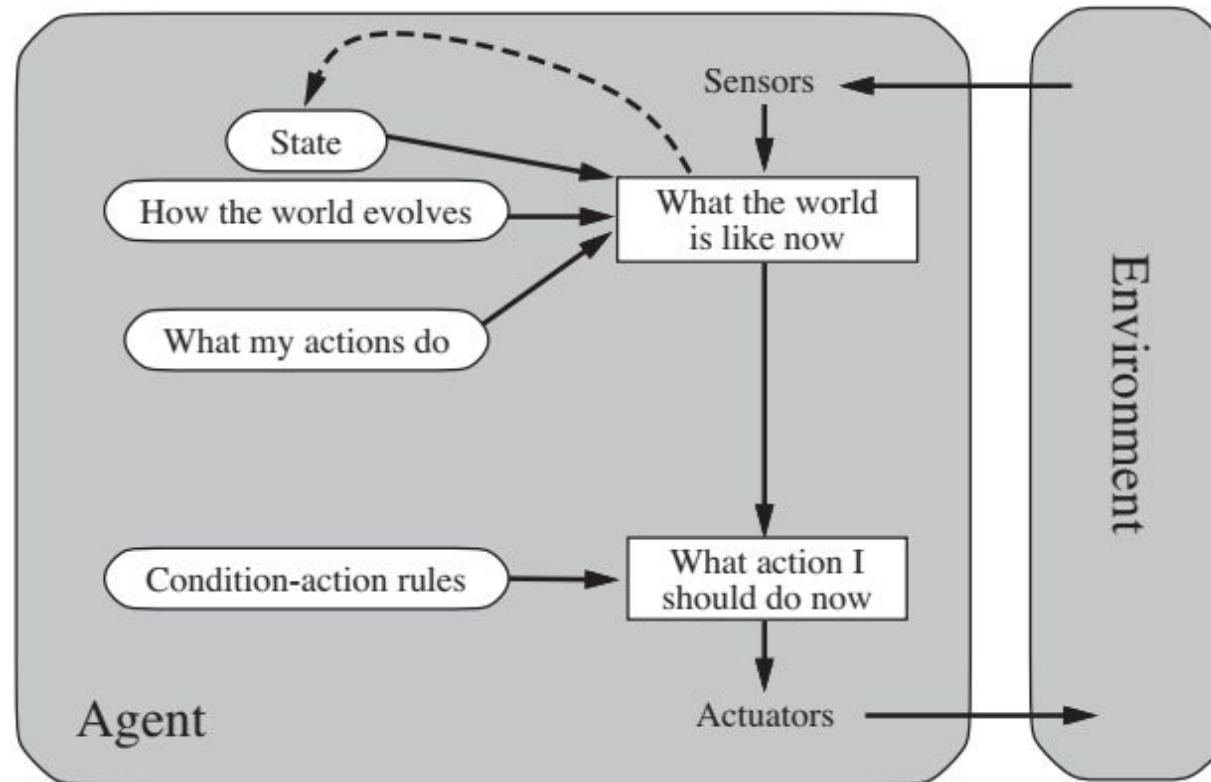
Types of Agent Programs

Model-Based Reflex Agents

- Self-driving cars are a great example of a model-based reflex agent.
- The car is equipped with sensors that detect
 - obstacles
 - Brake lights of vehicles in front of them (predicting their slowing down)
 - pedestrians walking on the sidewalk.
- As it drives, these sensors feed percepts into the car's memory and internal model of its environment.

Types of Agent Programs

Model-Based Reflex Agents



Types of Agent Programs

Goal-Based Agents

- A goal-based agent is an AI system that chooses its actions based on achieving a specific goal.
- Instead of just reacting to current conditions, these agents evaluate different possible actions and select the one that leads to a desired outcome.

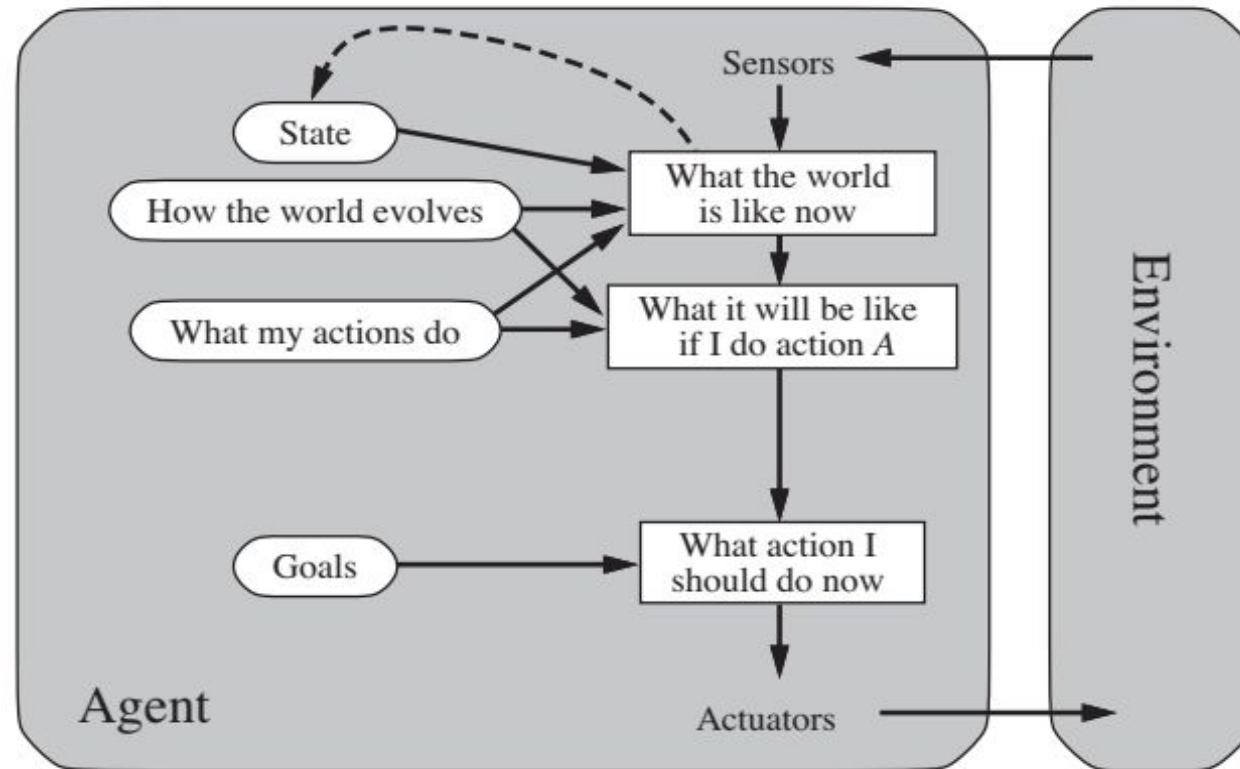
Types of Agent Programs

Goal-Based Agents

- Decision Making Based on Goals → The agent selects actions that will help achieve a predefined goal.
- Planning → Instead of acting immediately, the agent considers what will happen next after each action.
- Flexible Behavior → Can adapt its actions if conditions change, as long as the new path still leads to the goal.
- Requires Search → Often uses search algorithms (e.g., A* Search, BFS, DFS) and planning techniques to determine the best course of action.

Types of Agent Programs

Goal-Based Agents



Types of Agent Programs

Utility-Based Agents

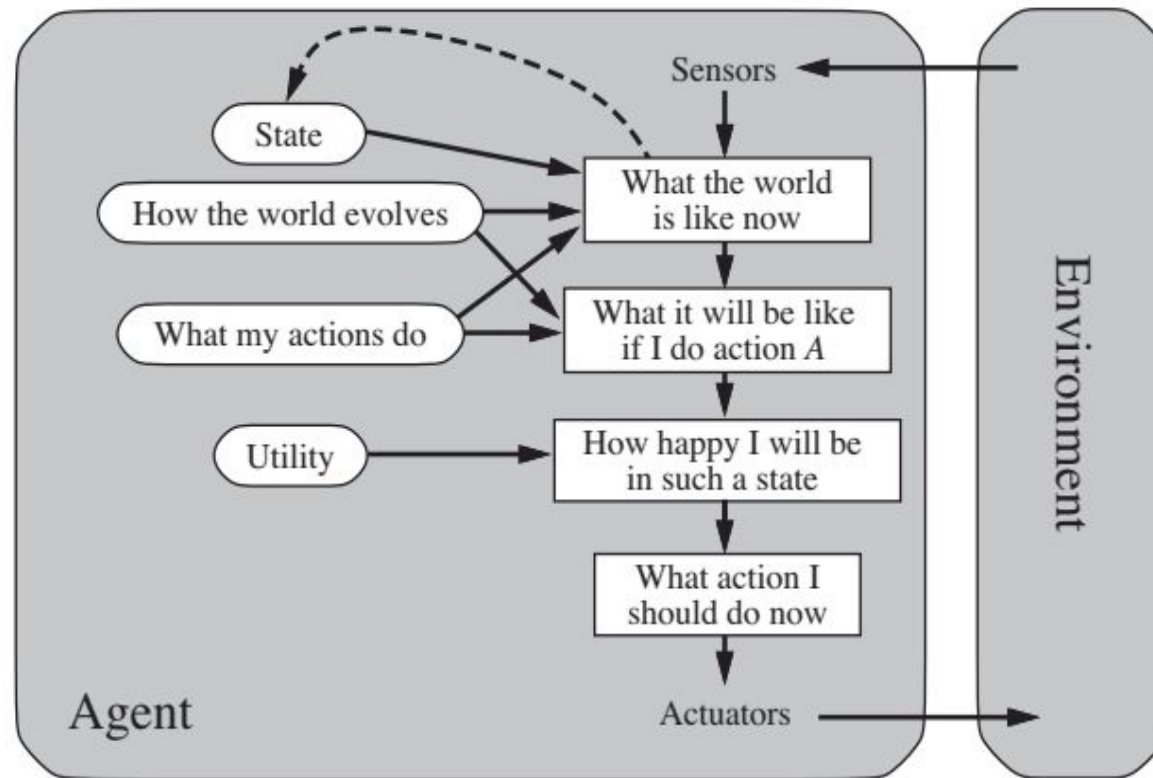
- Goals alone are not really enough to generate high-quality behavior in most environments.
- For example, there are many action sequences that will get the taxi to its destination (thereby achieving the goal) but some are quicker, safer, more reliable, or cheaper than others.

Types of Agent Programs

- A utility-based agent goes beyond just achieving a goal—it evaluates how beneficial each action is and chooses the best one based on a utility function.
- Example: Self-Driving Car as a Utility-Based Agent
- Goal: Reach Destination B from Point A.
- The agent assigns a utility score to different routes based on:
 - Time taken, Fuel efficiency, Traffic conditions, Comfort level.
- It picks the highest-utility route, balancing speed, fuel consumption, and safety.

Types of Agent Programs

Utility-Based Agents



Types of Agent Programs

- ***Learning Agent***

- A Learning Agent is an AI system that improves its performance over time by learning from past experiences.
- They improve over time: Other agents follow fixed rules, but learning agents adapt and optimize their decision-making.
- They use feedback: Unlike goal-based and utility-based agents, learning agents evaluate performance through a critic.
- They explore new possibilities: The problem generator allows learning agents to test new actions to improve.

Types of Agent Programs

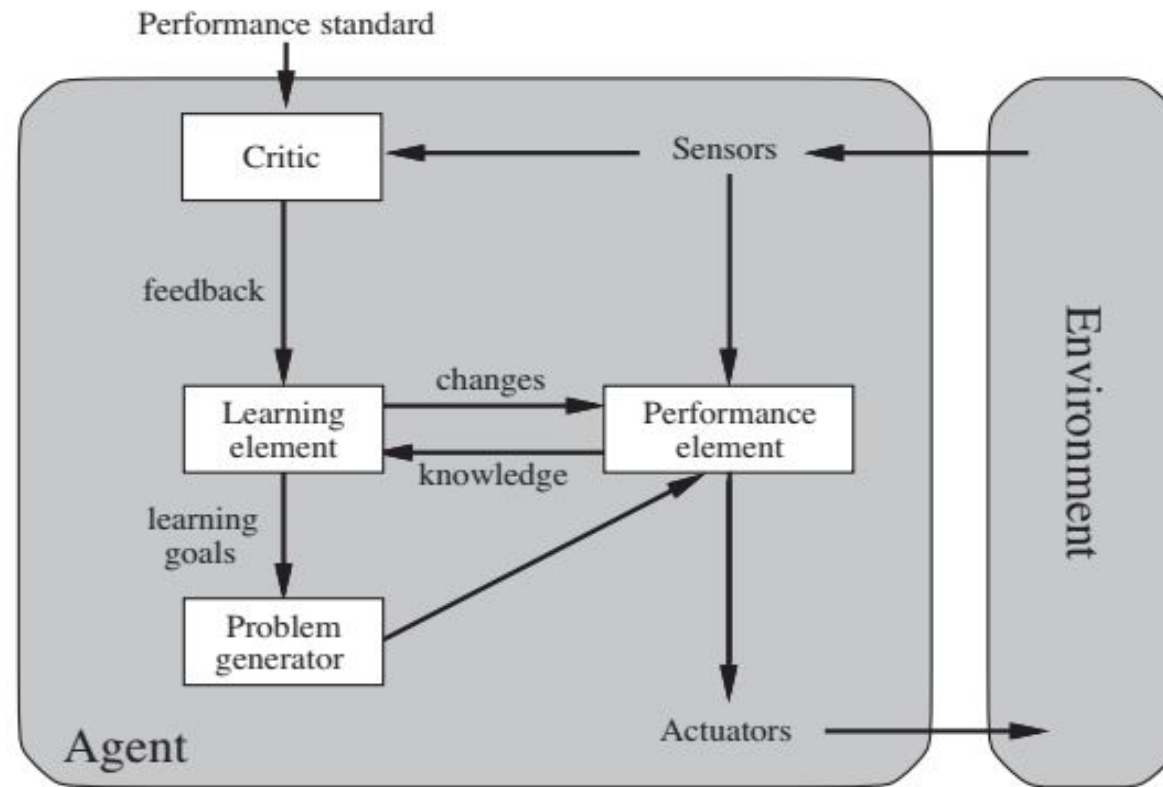
- **Learning Agent**

- A learning agent can be divided into four conceptual components, as:
 - **Learning element**- responsible for making improvements
 - **Performance element**- responsible for selecting external actions.
 - **Critic**- The learning element uses feedback from the critic on how the agent is doing and determines how the performance element should be modified to do better in the future.
 - **Problem generator**- and it is responsible for suggesting actions that will lead to new and informative experiences.

Types of Agent Programs

Learning Agent

- Schematic Diagram of Learning Agent is given below:



PEAS in AI (Performance Measure, Environment, Actuators, Sensors)

- PEAS is a framework used to **define and evaluate an AI agent's environment and functionality**.
- It helps in **designing intelligent agents** by clearly specifying **how they perceive the world, what actions they can take, and how their performance is measured**.
- It stands for:
 - **Performance Measure**: Output we get from the agent.
 - **Environment**: All surrounding things and conditions.
 - **Actuators**: Devices (hardware, software) through which agent performs actions on the environment
 - **Sensors**: Devices from which agent perceives information from the environment.

PEAS

- In designing an agent, the first step must always be to specify the task environment as fully as possible.
- The following table shows PEAS description of the task environment for a Medical Diagnosis System

Agent Type	Performance Measure	Environment	Actuators	Sensors
Medical diagnosis system	Healthy patient, reduced costs	Patient, hospital, staff	Display of questions, tests, diagnoses, treatments, referrals	Keyboard entry of symptoms, findings, patient's answers

Classification of Environment

Accessibility of Environment

- Is everything an agent requires to choose its actions available to it via its sensors?
 - If so, the environment is **accessible / fully observable**
- If not, parts of the environment are **inaccessible / partially observable**
 - Agent must make informed guesses about world

Determinism in the Environment

- Does the change in world state
 - Depend only on current state and agent's action?
- Non-deterministic environments
 - Have aspects beyond the control of the agent
 - Utility functions have to guess at changes in world

Static or Dynamic Environments

- **Static** environments don't change
 - While the agent is deliberating over what to do
- **Dynamic** environments do change
 - So agent should/could consult the world when choosing actions
 - Alternatively: anticipate the change during deliberation
 - Alternatively: make decision very fast

Discrete or Continuous Environments

- Nature of sensor readings / choices of action
 - Sweep through a range of values (continuous)
 - Limited to a distinct, clearly defined set (discrete)
- Maths in programs altered by type of data
- Chess: discrete; Genetic Systems: discrete

1. Classification of Environment in AI

Classification of Environment in AI:

→ Environment is part of the universe that surrounds Intelligent system.
→ Environment + Agent = Universe.

① Accessible and Inaccessible

- Agent can obtain complete and accurate information about state's environment. (Accessible) ←
- Whole part of the environment may not be in reach. (Inaccessible)
- Example of Accessible environment: Room with temperature as state.
- Example of Inaccessible environment: Any event on earth.

② Deterministic or Non-Deterministic

- Agent's current state and selected action can completely determine environment next state (Deterministic)
- Nature of environment can't be decided by agent alone (Non-deterministic/stochastic).
- Example of deterministic environment: Agent doesn't have to worry about uncertainty.
- Example of non-deterministic environment: Random in nature.

1. Classification of Environment in AI

③ Static or Dynamic

- ↳ Environment doesn't change its state with passage of time (Static).
- ↳ State of environment changes with respect to time (Dynamic)
- ↳ Example of static environment: crossword puzzles.
- ↳ " " dynamic " : Car driving.

④ Discrete or Continuous

- ↳ Finite no. of percepts and actions (Discrete), for example: Chess Game
- ↳ Indiscrete environment observations are not continuous.
- ↳ Agent can perceive and make observations from the environment continuously without any lag (Continuous), for example: Self-driving car

1. Classification of Environment in AI

⑤ Observable or Partially observable

↳ Agent sensor can access complete state of environment at each point of time.
(observable), It is fully or completely observable

↳ In partially observable, some part of environment is not in reach of agent.

References

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